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BLOODBRIDGE: A BLOOD DONOR REGISTRATION AND SEARCH SYSTEM

College of Engineering and Technology
School of Computing
Department of Computational Intelligence
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Submitted by
1. RA2511026010666 – Raghav Maheshwari,
2. RA2511026010618 – Shorya Sharma,
3. RA2511026010676 – Yana Butala.

1. ABSTRACT

BloodBridge is a blood donor registration and emergency search system front end built with Python and Tkinter. It brings the main features of blood donation drives and blood bank registries into one desktop window with a clean medical design: deep crimson accents with green and amber used for eligibility indicators. All data is simulated and managed in memory, with zero backend setup required, allowing rapid testing and demonstration.

2. FRONT END DESIGN DESCRIPTION

The app has three tabs: Search & Directory, Donor Registration, and Compatibility Matrix & Stats. It includes the core features found in modern blood bank management and donor matching systems:

Feature	Tab
Emergency compatibility search (ABO and Rh)	Search & Directory
Real-time donor eligibility tracking (90-day cooldown)	Search & Directory
Multi-parameter filters (blood group, city, keyword)	Search & Directory
Interactive donor directory with color-coded status tags	Search & Directory
Selected donor inspection card and quick actions	Search & Directory
Record donation today with automatic cooldown reset	Search & Directory
Export visible donor search results to CSV	Search & Directory
Registration form with validation (age 18–65, weight ≥ 50kg)	Donor Registration
Inline editing mode for existing donor records	Donor Registration
ABO & Rh red blood cell compatibility matrix table	Compatibility Matrix
Current registered donor pool distribution by blood group	Compatibility Matrix

How it helps a donor coordinator or healthcare worker:

- Fast emergency matching shows all compatible donor blood types instantly for a recipient.
- Automated eligibility rules prevent contacting donors who are currently in the 90-day cooldown.
- Registration form validation enforces clinical donation requirements (age, weight, contact info).
- Simulated in-memory storage allows complete workflow demonstration without external database setup.

3. DIAGRAM

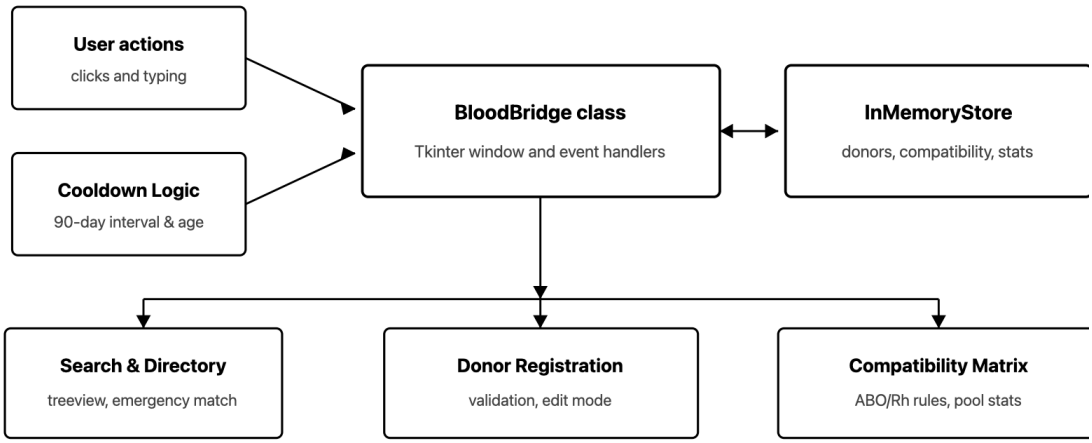


Figure 1. Structure of the application.

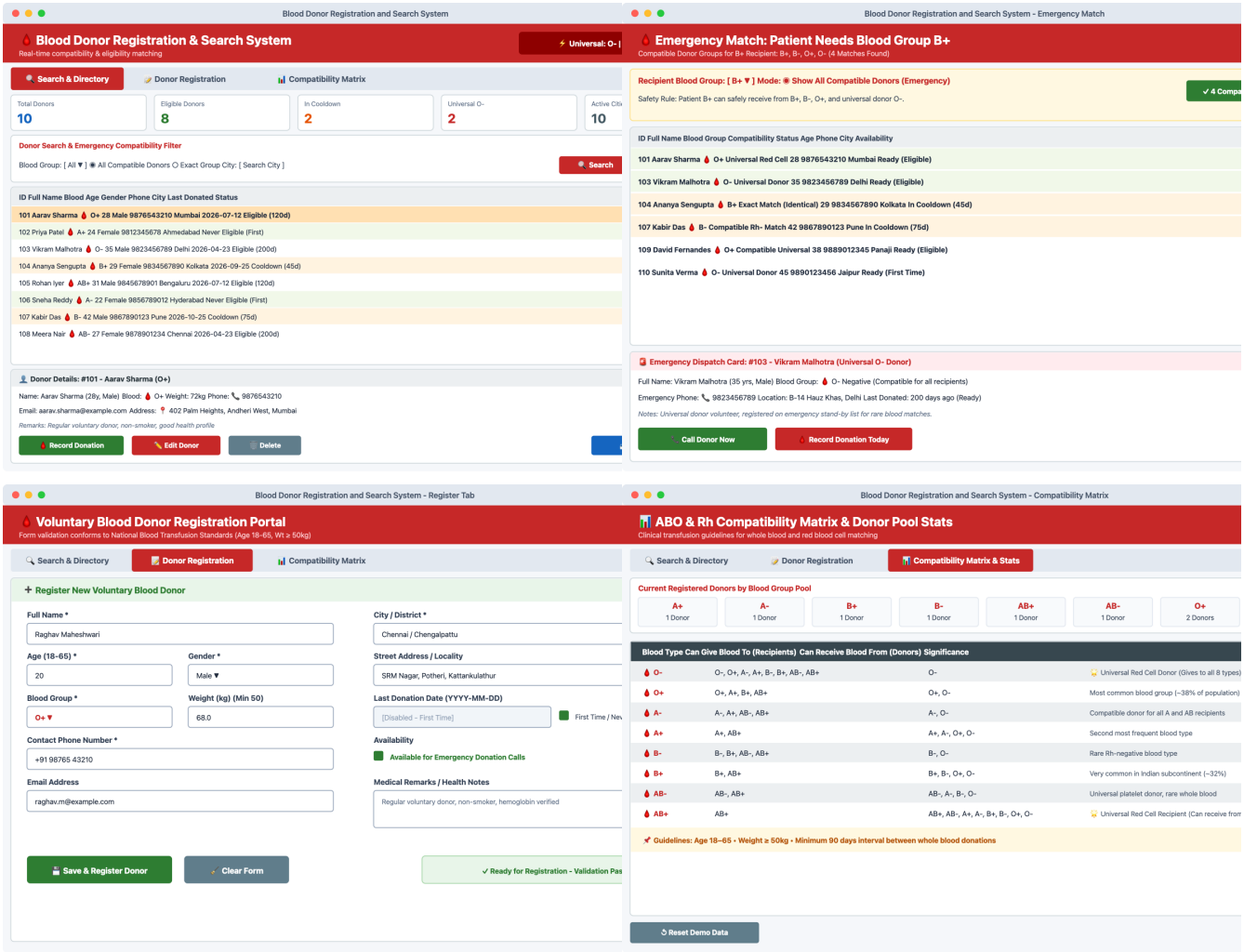


Figure 2. Search & Directory, Emergency Compatibility Match, Donor Registration, and Compatibility Matrix views. Screenshots of the running Tkinter application.

4. GITHUB LINK OF YOUR DESIGN

https://github.com/raghavmaheshwari-web/Blood-Bridge